

Lab 2 — What I ran, what I saw, what I would ship

1. What I was trying to do

- Lab 1 gave me a working extractor. Lab 2 asked: is this the best version, or just the one I happened to try?
- I ran a grid: zero-shot / few-shot / few-shot+reasoning × SMALL / MAIN, plus a cascade (SMALL first, MAIN if unsure).
- I measured quality, cost, and latency together — not accuracy alone.
- I used paired tests so a 3–5 point bump on 60 tickets would not get over-read.

2. Grid observations (what the table actually showed)

- **zero_shot SMALL:** record ~0.30, field ~0.87, schema 1.0. Cheap baseline; fully reliable when not rate-limited.
- **few_shot SMALL:** record ~0.39, field ~0.88. Point estimate higher; paired vs zero_shot $p=0.42$ → not significant.
- **few_shot reasoned SMALL:** record ~0.43, field ~0.90. Best point estimates; paired $p=0.29$ → still noise. Extra tokens, no clear win.
- **MAIN variants:** mostly failed (90–100% errors) on free-tier daily quota. Not a fair model comparison — infrastructure failure, not “MAIN is worse”.
- **cascade:** escalation rate 0%. Matched zero_shot on every item. My trigger (failed validation or evidence < 12 chars) never fired on the completed run.

3. Error patterns I kept seeing (confusion-style)

I did not get a full printed confusion matrix from the harness, but reading failures and per-field scores, the mistakes clustered the same way every time:

- **Urgency (worst field):** off-by-one at the official boundaries. $1 \leftrightarrow 2$ (can this be answered without opening the customer record?) and $2 \leftrightarrow 3$ (already stuck vs first-time defect). Not random scatter — systematic boundary confusion.

Possible reasons: the model reacts to tone/length; the gold scale is situation-based. Shouting is sentiment, not urgency — easy to mix up without the decision question in front of the model every time.

- **Category — complaint vs concrete:** angry claims/billing tickets often pulled toward complaint. Gold rule: prefer the concrete category if the customer still wants that issue fixed; complaint only when Aurora’s conduct is the subject.

Possible reasons: anger is a strong surface signal; the boundary is subtle and needs explicit “prefer concrete” pressure in the schema/prompt.

- **Sentiment — neutral vs frustrated:** first-time terse requests sometimes marked frustrated; tickets that mention prior delay sometimes left neutral.

Possible reasons: gold requires frustrated to reference a prior failure. Without that anchor, any complaint-shaped language looks frustrated.

- **Deterministic fields (policy_number, product, language):** stayed near-perfect once moved out of the model in Lab 1. Errors here were rare and usually data-edge, not judgement.

4. What surprised me / negative results

- Few-shot did not “obviously” win once I ran the paired test. Higher raw accuracy is not the same as a deployable improvement.
- Reasoning field: more tokens, no significant accuracy gain on this set.
- Cascade with a weak free trigger can look sophisticated and still be pure SMALL (escalation 0%).
- Rate limits can fake a quality story if you do not separate “model wrong” from “call never happened.”

5. Recommendation (from my perspective)

- **Ship zero_shot on SMALL** (Lab 1 Part C: model only for judgement fields; code for policy number, PII, escalate).
- Reason: nothing beat it with statistical support; cost and simplicity win.
- I would change my mind if: a clean MAIN run, or a cascade with a stronger disagreement trigger (e.g. two SMALL samples at $T>0$), showed a paired-significant accuracy gain large enough to pay for the extra cost.

Bottom line: the harness mattered more than any single prompt. Once the numbers were honest, the shipping decision was mostly “choose on cost.”